

```

00001 *****
00002 ** DL Register Area - 120Word
00003 **
00004 ** DW00000 - DW00009 (Pls Use)
00005 ** DW00010 - DW00039 (Bit Use)
00006 ** DW00040 - DW00089 (Word Use)
00007 ** DW00090 - DW00119 (Timer)
00008 *****
00009
00010 "# Program Name      : STR1 Cycle 10 (V.S/PreAlignment)
00011 "# Function          :
00012 "# Call Program      : MPS481
00013 "# Sub Program       :
00014 "# Func Program      :
00015
00016 VAR;
00017     LONG STR_L_CAM_MARK_I           %ML97300;
00018     LONG STR_R_CAM_MARK_I           %ML97330;
00019
00020     BOOL STR_ALIGN_L_REQ             %MB970004;
00021     BOOL STR_ALIGN_R_REQ             %MB970005;
00022
00023     BOOL STR_L_CAM_LINE_RE           %MB970024;
00024     BOOL STR_R_CAM_LINE_RE           %MB970025;
00025
00026     BOOL STR_ALIGN_L_MANUAL_RE        %MB970044;
00027     BOOL STR_ALIGN_R_MANUAL_RE        %MB970045;
00028
00029     BOOL STR_L_CAM_DL_ALIGN_RE        %MB970084;
00030     BOOL STR_R_CAM_DL_ALIGN_RE        %MB970085;
00031
00032     BOOL STR_ALIGN_L_RESP             %MB970104;
00033     BOOL STR_ALIGN_R_RESP             %MB970105;
00034
00035     BOOL STR_L_CAM_LINE_RE           %MB970124;
00036     BOOL STR_R_CAM_LINE_RE           %MB970125;
00037
00038     BOOL STR_L_CAM_MANUAL_ALIGN_RE    %MB970144;
00039     BOOL STR_R_CAM_MANUAL_ALIGN_RE    %MB970145;
00040
00041     BOOL STR_L_CAM_DL_ALIGN_RE        %MB970184;
00042     BOOL STR_R_CAM_DL_ALIGN_RE        %MB970185;
00043
00044     BOOL STR_L_CAM_ALIGN_RESULT_C     %MB970600;
00045     BOOL STR_R_CAM_ALIGN_RESULT_C     %MB970700;
00046
00047     BOOL STR_L_CAM_ALIGN_RESULT_I     %MB970601;
00048     BOOL STR_R_CAM_ALIGN_RESULT_I     %MB970701;
00049
00050     LONG STR_L_CAM_ALIGN_ERROR_COI    %ML97062;      // 0:OK 1:NO MARK ID 2:MANUAL SEARCH 3:MANUA
00051 L OK 4:MANUAL ABORT
00052     LONG STR_R_CAM_ALIGN_ERROR_COI    %ML97072;      // 0:OK 1:NO MARK ID 2:MANUAL SEARCH 3:MANUA
00053 L OK 4:MANUAL ABORT
00054     BOOL STR_L_CAM_MANUAL_ALIGN_IN    %MB970622;
00055     BOOL STR_R_CAM_MANUAL_ALIGN_IN    %MB970722;
00056
00057     BOOL STR_L_CAM_MANUAL_ALIGN_ABOI  %MB970624;
00058     BOOL STR_R_CAM_MANUAL_ALIGN_ABOI  %MB970724;
00059
00060     LONG STR_L_CAM_ALIGN_SEARCH_TYI   %ML97314;      // 0:Mark Search 1:Edge Search 2:DL 3:Manual
00061 4:Auto Mark Search
00062     LONG STR_R_CAM_ALIGN_SEARCH_TYI   %ML97344;
00063
00064     BOOL STR_ALIGN_MANUALSTOP_ALARM    %MB075381;
00065     BOOL STR_ALIGN_MARK_DISTANCE_ALAI %MB075382;
00066     BOOL STR_ALIGN_ZERO_VAL_ALARM      %MB075383;
00067
00068     LONG STR_ALIGN_L_RESULT_XDAI       %ML97064;
00069     LONG STR_ALIGN_L_RESULT_YDAI       %ML97066;
00070     LONG STR_ALIGN_R_RESULT_XDAI       %ML97074;
00071     LONG STR_ALIGN_R_RESULT_YDAI       %ML97076;
00072
00073 END_VAR;
00074 BLK MW24160 MW97304 W8;
00075 BLK MW24160 MW97334 W8;
00076
00077 CLR DW00050 W20;
00078 ML07892 = 0;      " PreAlign Auto Mark Search X Of
00079 fset Result
00080 ML07894 = 0;      " PreAlign Auto Mark Search Y Of
00081 fset Result
00082
00083 STR_ALIGN_L_REQ = 0      " Alignment Request OFF(Left)
00084 STR_ALIGN_R_REQ = 0      " Alignment Request OFF(Right)
00085 STR_L_CAM_LINE_REQ = 0;
00086 STR_R_CAM_LINE_REQ = 0;

```

```

00083 00009      STR_ALIGN_L_MANUAL_REQ = 0;
00084 00010      STR_ALIGN_R_MANUAL_REQ = 0;
00085 00011      STR_L_CAM_DL_ALIGN_REQ = 0;
00086 00012      STR_R_CAM_DL_ALIGN_REQ = 0;
00087
00088 00013      STR_L_CAM_MARK_ID = GL51018;
00089 00014      STR_R_CAM_MARK_ID = GL51018;
00090
00091 00015      IF !MB076323 == 1                                " PreAlignment Po
00092 00016          MSEE MPS480;
00093 00017          TIM TMW30430;
00094 00018      IEND;
00095
00096
00097 00019      IOW STR_ALIGN_L_RESP | STR_ALIGN_R_RESP == 0;
00098
00099      // IF !MB000800 == 1                                " No-Load Operatio
00100 00020          DB00020 = 0;
00101 00021          DB00021 = 0;
00102 00022          DW00020 = 0;
00103
00104 00023      WHILE DW00020 < MW30440;
00105 00024          STR_L_CAM_ALIGN_SEARCH_TYPE = 4;
00106 00025          STR_R_CAM_ALIGN_SEARCH_TYPE = 4;
00107
00108 00026          STR_ALIGN_L_REQ = 1                                " Alignment Request ON(Left
00109 00027          STR_ALIGN_R_REQ = 1                                " Alignment Request ON(Right)
00110 00028          IOW STR_L_CAM_ALIGN_ERROR_CODE == 0;
00111 00029          IOW STR_R_CAM_ALIGN_ERROR_CODE == 0;
00112
00113 00030      IOW STR_ALIGN_L_RESP & STR_ALIGN_R_RESP==                " Alignment Respons O
00114
00115 00031      IF STR_L_CAM_ALIGN_RESULT_OK == 1
00116 00032          ML07810 = STR_ALIGN_L_RESULT_XDATA;                " Alignment Data X1
00117 00033          ML07812 = STR_ALIGN_L_RESULT_YDATA * -                " Alignment Data Y1
00118 00034          DB00020 = 1;
00119 00035      IEND;
00120 00036      IF STR_R_CAM_ALIGN_RESULT_OK == 1;
00121 00037          ML07814 = STR_ALIGN_R_RESULT_XDATA;                " Alignment Data X2
00122 00038          ML07816 = STR_ALIGN_R_RESULT_YDATA * -                " Alignment Data Y2
00123 00039          DB00021 = 1;
00124 00040      IEND;
00125
00126 00041      MB07860A = STR_L_CAM_ALIGN_RESULT_NG;
00127 00042      MB07860B = STR_L_CAM_ALIGN_RESULT_NG;
00128
00129 00043      STR_ALIGN_L_REQ = 0                                " Alignment Request OFF(Left)
00130
00131 00044      STR_ALIGN_R_REQ = 0                                " Alignment Request OFF(Right)
00132
00131
00132 00045      IF DB00020 & DB00021 == 1;
00133 00046          DW00020 = MW30440 + 1;
00134 00047          IF CW00054 == 1;
00135 00048              MB07860A = 0;
00136 00049              MB07860B = 0;
00137 00050          IEND;
00138 00051      ELSE;
00139 00052          DW00020 = DW00020 + 1;
00140 00053          TIM T100;
00141 00054      IEND;
00142 00055      WEND;
00143 00056      IF GL51020 == 0                                " Auto Mark Search (0:Use 1:Unus
e)
00144 00057          IF (MB07860A | MB07860B) == 1;
00145 00058              IF MB078606 == 0;                                " Auto Mark Seach Complete
00146 00059                  MSEE MPS485;
00147 00060              IEND;
00148 00061          IEND;
00149 00062      IEND;
00150
00151      " Manual Search Flag
00152 00063      IF MB07860A | MB07860B == 1;
00153
00154 00064          DB000000 = 0;
00155 00065          DB000010 = 0;
00156 00066          DB000011 = 0;
00157 00067          WHILE DB000000 == 0
00158 00068              STR_L_CAM_ALIGN_SEARCH_TYPE = 0;
00159 00069              STR_R_CAM_ALIGN_SEARCH_TYPE = 0;
00160
00161 00070          IOW STR_ALIGN_L_RESP | STR_ALIGN_R_RESP ==                " Wait Alignment Respons OFF
00162 00071          STR_ALIGN_L_REQ = MB07860A;                        " L Camera Manual Alignment
Request
00163 00072          STR_ALIGN_R_REQ = MB07860E;                        " R Camera Manual Alignment

```

```

Request
00164 00073      EOX;
00165
00166 00074      IOW (STR_L_CAM_ALIGN_RESULT_OK | STR_L_CAM_MANUAL_ALIGN_ING) & (STR_R_CAM_ALIGN_RESU
LT_OK | STR_R_CAM_MANUAL_ALIGN_ING) ==      " Alignment Respons O
00167
00168 00075      IF STR_L_CAM_ALIGN_RESULT_OK & STR_R_CAM_ALIGN_RESULT_OK == 1;
00169
00170 00076      ELSE;
00171 00077          PFORK Line1 Line2;
00172          Line1:
00173 00078              IF STR_L_CAM_MANUAL_ALIGN_ING == 1;
00174 00079                  IOW !STR_L_CAM_MANUAL_ALIGN_ING == 1;
00175 00080                  IOW STR_ALIGN_L_RESP == 1;
00176 00081                  IEND;
00177 00082                  JOINTO LabelX1;
00178
00179          Line2:
00180 00083              IF STR_R_CAM_MANUAL_ALIGN_ING == 1;
00181 00084                  IOW !STR_R_CAM_MANUAL_ALIGN_ING == 1;
00182 00085                  IOW STR_ALIGN_R_RESP == 1;
00183 00086                  IEND;
00184 00087                  JOINTO LabelX1;
00185 00088          LabelX1: PJOINT;
00186
00187 00089          IF STR_L_CAM_MANUAL_ALIGN_ABORT == 1;
00188 00090              MB000015 = 1;      " Cycle Stop
00189 00091              STR_ALIGN_MANUALSTOP_ALARM = 1      " Alignment Resp NG
00190 00092              STR_ALIGN_L_REQ = 0;
00191 00093              DB000000 = 1;
00192 00094              EOX;
00193 00095          IEND;
00194 00096          IF STR_R_CAM_MANUAL_ALIGN_ABORT == 1;
00195 00097              MB000015 = 1;      " Cycle Stop
00196 00098              STR_ALIGN_MANUALSTOP_ALARM = 1      " Alignment Resp NG
00197 00099              STR_ALIGN_R_REQ = 0;
00198 00100              DB000000 = 1;
00199 00101              EOX;
00200 00102          IEND;
00201 00103      IEND;
00202
00203 00104      IF STR_L_CAM_ALIGN_RESULT_OK == 1;
00204 00105          ML07810 = STR_ALIGN_L_RESULT_XDATA + (IL8C16 - ML6480)      " Alignm
ent Data X1
00205 00106          ML07812 = STR_ALIGN_L_RESULT_YDATA * -1 - (IL8816 - ML6322      " Alignm
ent Data Y1
00206 00107          MB07860A = 0;      " L Alig
nment NG (0:OK 1:NG)
00207 00108          STR_ALIGN_L_MANUAL_REQ = 0      " L Came
ra Manual Alignment Request
00208 00109          IEND;
00209
00210 00110      IF STR_R_CAM_ALIGN_RESULT_OK == 1;
00211 00111          ML07814 = STR_ALIGN_R_RESULT_XDATA + (IL8C96 - ML6500)      " Alignm
ent Data X2
00212 00112          ML07816 = STR_ALIGN_R_RESULT_YDATA * -1 - (IL8816 - ML6322      " Alignm
ent Data Y2
00213 00113          MB07860B = 0;      " R Alig
nment NG (0:OK 1:NG)
00214 00114          STR_ALIGN_R_MANUAL_REQ = 0      " R Came
ra Manual Alignment Request
00215 00115          IEND;
00216
00217 00116          STR_ALIGN_L_REQ = 0      " L Came
ra Manual Alignment Request OFF
00218 00117          STR_ALIGN_R_REQ = 0      " R Came
ra Manual Alignment Request OFF
00219 00118          IOW STR_ALIGN_L_RESP | STR_ALIGN_R_RESP ==      " Wait A
lignment Respons OFF
00220
00221 00119          IF MB07860A | MB07860B == 0;
00222 00120              DB000000 = 1;
00223 00121          IEND;
00224 00122          EOX;
00225 00123          WEND;
00226 00124      IEND;
00227 00125      IF !MB000015 == 1      " Cycle Stop
00228 00126          IF ML07810 == 0      " Alignment ZeroValue Check
00229 00127              IF ML07812 == 0;
00230 00128                  IF ML07814 == 0      " Alignment ZeroValue Check
00231 00129                      IF ML07816 == 0;
00232 00130                          STR_ALIGN_ZERO_VAL_ALARM = 1      " Alignment ZeroValue Alarm
00233 00131                          EOX;
00234 00132                      IEND;
00235 00133                  IEND;
00236 00134              IEND;
00237 00135          IEND;

```

```
00238
00239 00136          DF00050 = IL8C16 + ML0781C          " Left Mark Real X Po

00240 00137          DF00052 = IL8C96 + ML07814;          " Right Mark Real X Pos
00241 00138          DF00054 = ML07812 - ML07816;          " L,R Mark Y Distance
00242 00139          DL00056 = GL51016          " Align Mark Pos D

00243 00140          DF00058 = DF00052 - DF00050;

00244
00245 00141          DF00060 = (DF00058 * DF00058) + (DF00054 * DF00054);
00246 00142          DF00062 = SQT(DF00060);
00247 00143          DL00064 = DL00056 - DF00062;

00248
00249 00144          IF DL00064 < 0,          " Find Mark Distance Check
00250 00145              DL00064 = -DL00064;
00251 00146          IEND;

00252
00253 00147          IF DL00064 > ML30434;
00254 00148              STR_ALIGN_MARK_DISTANCE_ALARM = 1;
00255 00149          IEND;
00256
00257 00150          IEND;

00258
00259 00151          MSEE MPS465;          " Stick Transfer 1 Axis Settir
00260          // ELSE;
00261          //          ML07810 = 0;          " Alignment Data X1
00262          //          ML07812 = 0;          " Alignment Data Y1
00263          //          ML07814 = 0;          " Alignment Data X2
00264          //          ML07816 = 0;          " Alignment Data Y
00265          // IEND;
00266
00267 00152          RET;
```